

Mathematics for Data Science 2

Linear Algebra & Multivariable Calculus for Data Science

IIT Madras BS Degree Programme

IITM BS Degree Curriculum Team

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Chapter 1

Vectors, Matrices & Determinants

Welcome to Mathematics for Data Science 2! Linear algebra is the foundational mathematical backbone of modern Machine Learning, Artificial Intelligence, and Data Science. From storing high-dimensional feature spaces to computing neural network transformations and performing dimensionality reduction, vectors and matrices are the fundamental building blocks.

1.1 Vectors in $\mathbb{R}^n$

A **Vector** in n -dimensional real space $\mathbb{R}^n$ is an ordered sequence of n real numbers, written as a column matrix:

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{R}^n$$

Definition: Vector Operations

For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and scalar $c \in \mathbb{R}$:

- **Vector Addition:** $\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{bmatrix}$
- **Scalar Multiplication:** $c\mathbf{v} = \begin{bmatrix} cv_1 \\ cv_2 \\ \vdots \\ cv_n \end{bmatrix}$
- **Dot Product:** $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = \sum_{i=1}^n u_i v_i = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$

Data Science Relevance: In data science, a sample (e.g. a user profile with age, income, credit score) is represented as a feature vector $\mathbf{x} \in \mathbb{R}^d$. The linear combination $\mathbf{w}^T \mathbf{x} + b$ forms the prediction score in Linear and Logistic Regression!

Video Resource: IITM BS Lecture: Vectors 

Official lecture defining vector spaces, geometric interpretations, and algebraic operations in $\mathbb{R}^n$.

1.2 Matrices and Matrix Operations

A **Matrix** $A \in \mathbb{R}^{m \times n}$ is a rectangular array of m rows and n columns:

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

Theorem / Rule: Matrix Multiplication

If $A \in \mathbb{R}^{m \times k}$ and $B \in \mathbb{R}^{k \times n}$, their product $C = AB \in \mathbb{R}^{m \times n}$ has entries:

$$c_{ij} = \sum_{p=1}^k a_{ip} b_{pj}$$

Matrix multiplication is associative $(AB)C = A(BC)$ and distributive, but **NOT commutative** ($AB \neq BA$ in general).

Video Resource: IITM BS Lecture: Matrices 

Official lecture explaining matrix algebra, transposition A^T , symmetric matrices ($A = A^T$), and matrix-vector products.

1.3 Systems of Linear Equations

A system of m linear equations in n variables $x_1, x_2, \dots, x_n$ can be compactly written in matrix form:

$$A\mathbf{x} = \mathbf{b}$$

where $A \in \mathbb{R}^{m \times n}$ is the coefficient matrix, $\mathbf{x} \in \mathbb{R}^n$ is the vector of unknowns, and $\mathbf{b} \in \mathbb{R}^m$ is the target output.

Definition: Types of Solutions

Every linear system $A\mathbf{x} = \mathbf{b}$ has either:

1. Exactly **one unique solution** (consistent, non-singular).
2. **Infinitely many solutions** (consistent, dependent).
3. **No solution** (inconsistent, parallel constraints).

Video Resource: IITM BS Lecture: Systems of Linear Equations 

Official lecture introducing linear systems, augmented matrices $[A \mid \mathbf{b}]$, and geometric interpretations of solutions.

1.4 Determinants

The **Determinant** $\det(A)$ or $|A|$ is a scalar value associated with a square matrix $A \in \mathbb{R}^{n \times n}$ that measures the scaling factor of transformation area/volume.

Key Formula / Algorithm: Determinant Computation (2×2 and 3×3)

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$:

$$\det(A) = ad - bc$$

For $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$, expanding along the first row:

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg)$$

Invertibility Criterion: Matrix A is invertible (A^{-1} exists) if and only if $\det(A) \neq 0$.

Video Resource: IITM BS Lecture: Determinants (Parts 1 & 2) 

Official lectures explaining cofactor expansion, properties of determinants, and invertibility conditions.

1.5 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Vector Dot Product & Angle:** Given vectors $\mathbf{u} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}^T$:
- (a) Compute the dot product $\mathbf{u} \cdot \mathbf{v}$.
 - (b) Compute the norms $\|\mathbf{u}\|$ and $\|\mathbf{v}\|$.

2. **Matrix Multiplication:** Given matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & -1 \\ 2 & 5 \end{bmatrix}$$

Compute AB and BA . Verify that $AB \neq BA$.

3. **Matrix-Vector System Formulation:** Convert the linear system:

$$\begin{aligned} 2x_1 + 3x_2 - x_3 &= 7 \\ 4x_1 - x_2 + 2x_3 &= 3 \\ -x_1 + 5x_2 + 3x_3 &= 10 \end{aligned}$$

into the matrix form $A\mathbf{x} = \mathbf{b}$ and write down the augmented matrix $[A \mid \mathbf{b}]$.

4. **Determinant Calculation:** Calculate the determinant of the 3×3 matrix:

$$M = \begin{bmatrix} 3 & 1 & 2 \\ 0 & 4 & 5 \\ 1 & 2 & 1 \end{bmatrix}$$

Is matrix M invertible?

5. **Data Science Application (Covariance Matrix Determinant):** In multivariate statistics, the determinant of a 2×2 Covariance Matrix $\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix}$ represents total generalized variance. Given $\sigma_1^2 = 4$, $\sigma_2^2 = 9$, and covariance $\sigma_{12} = 3$, calculate $\det(\Sigma)$.

Chapter 2

Cramer's Rule, Row Echelon Form & Gaussian Elimination

Solving large systems of linear equations $Ax = b$ is the single most common computational operation in numerical algorithms. In this chapter, we master explicit solutions via Cramer's Rule and systematic algorithmic reduction via **Gaussian Elimination** and **Row Echelon Form (REF)**.

2.1 Cramer's Rule

For a square, non-singular linear system $Ax = b$ with $\det(A) \neq 0$, **Cramer's Rule** gives an explicit formula for each variable x_i using determinants.

Theorem / Rule: Cramer's Rule Formula

Let $A_i(b)$ be the matrix obtained by replacing the i -th column of A with the constant vector b . The unique solution for variable x_i is:

$$x_i = \frac{\det(A_i(b))}{\det(A)}, \quad \text{for } i = 1, 2, \dots, n$$

Example: Solving 2×2 System via Cramer's Rule

$$\text{Solve } \begin{cases} 2x + 3y = 8 \\ 5x - y = 3 \end{cases} :$$

$$\det(A) = \begin{vmatrix} 2 & 3 \\ 5 & -1 \end{vmatrix} = -2 - 15 = -17$$

$$\det(A_x(b)) = \begin{vmatrix} 8 & 3 \\ 3 & -1 \end{vmatrix} = -8 - 9 = -17 \implies x = \frac{-17}{-17} = 1$$

$$\det(A_y(b)) = \begin{vmatrix} 2 & 8 \\ 5 & 3 \end{vmatrix} = 6 - 40 = -34 \implies y = \frac{-34}{-17} = 2$$

Video Resource: IITM BS Lecture: Cramer's Rule 

Official lecture deriving Cramer's Rule and applying it to non-singular linear systems.

2.2 Elementary Row Operations & Row Echelon Form (REF)

While Cramer's Rule is theoretically elegant, calculating determinants for large $n \times n$ matrices is computationally intractable ($\mathcal{O}(n!)$ time). Instead, we transform augmented matrices using **Elementary Row Operations (EROs)** without changing the solution set.

Definition: Elementary Row Operations (EROs)

1. **Row Swap** ($R_i \leftrightarrow R_j$): Swap two rows.
2. **Scalar Multiplication** ($R_i \rightarrow cR_i$): Multiply a row by a non-zero scalar $c \neq 0$.
3. **Row Addition** ($R_i \rightarrow R_i + cR_j$): Add a scalar multiple of row j to row i .

Definition: Row Echelon Form (REF) & Reduced REF (RREF)

A matrix is in **Row Echelon Form (REF)** if:

1. All zero rows are at the bottom of the matrix.
2. The leading non-zero entry (pivot) of a non-zero row is strictly to the right of the pivot in the row above it.
3. All entries below a pivot are zero.

It is in **Reduced Row Echelon Form (RREF)** if, additionally, every pivot entry is 1, and all entries above each pivot are zero.

Video Resource: IITM BS Lecture: Row Echelon Form



Official lecture defining pivot entries, REF conditions, and RREF canonical form.

Video Resource: IITM BS Lecture: Row Reduction Techniques



Official lecture performing row operations step-by-step on rectangular matrices.

2.3 Gaussian Elimination & Back-Substitution

Gaussian Elimination reduces an augmented matrix $[A \mid \mathbf{b}]$ into REF using EROs, followed by **Back-Substitution** to solve for variables.

Key Formula / Algorithm: Gaussian Elimination Algorithm ($\mathcal{O}(n^3)$)

1. **Forward Elimination:** Use EROs to create zeros below each pivot from left to right, producing REF.
2. **Back-Substitution:** Starting from the bottom row, solve for each basic variable and substitute upwards.

Data Science Relevance: Gaussian Elimination and LU Decomposition under grid computing power the solver engines of NumPy ('np.linalg.solve'), SciPy, and Linear Programming solvers.

Video Resource: IITM BS Lecture: The Gaussian Elimination Method 

Official lecture executing full Gaussian elimination and identifying free vs pivot variables.

2.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Cramer's Rule (2×2):** Use Cramer's Rule to solve the system:

$$\begin{aligned} 3x_1 + 4x_2 &= 10 \\ x_1 - 2x_2 &= -3 \end{aligned}$$

2. **Identifying REF:** State whether each of the following matrices is in REF, RREF, or neither:

$$(a) A = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{bmatrix}, \quad (b) B = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}, \quad (c) C = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \end{bmatrix}$$

3. **Gaussian Elimination (3×3):** Solve the linear system using Gaussian Elimination (forward elimination to REF + back-substitution):

$$\begin{aligned} x_1 + 2x_2 + x_3 &= 8 \\ 2x_1 + 3x_2 + 4x_3 &= 20 \\ 4x_1 + 3x_2 + 2x_3 &= 16 \end{aligned}$$

4. **Free Variables & Infinite Solutions:** Reduce the augmented matrix to REF and write the general parametric solution in terms of free variables:

$$\begin{aligned} x_1 + 2x_2 - x_3 &= 4 \\ 2x_1 + 4x_2 + x_3 &= 11 \end{aligned}$$

5. **Data Science Application (Least Squares Normal Equations):** In linear regression, the parameters $\mathbf{w}$ are found by solving $(X^T X)\mathbf{w} = X^T \mathbf{y}$. Given:

$$X^T X = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}, \quad X^T \mathbf{y} = \begin{bmatrix} 10 \\ 9 \end{bmatrix}$$

Solve for parameter vector $\mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix}$ using Gaussian Elimination or Cramer's Rule.

Chapter 3

Vector Spaces, Subspaces & Linear Independence

Abstract algebra generalizes $\mathbb{R}^n$ into the formal concept of a **Vector Space**. Understanding vector spaces and linear independence is crucial for recognizing feature redundancies and understanding high-dimensional data representations.

3.1 Definition of a Vector Space

Definition: Vector Space Axioms

A **Vector Space** $(V, +, \cdot)$ over a field $\mathbb{R}$ is a set V equipped with vector addition $+$ and scalar multiplication $\cdot$ satisfying 8 fundamental axioms:

1. **Associativity of Addition:** $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
2. **Commutativity of Addition:** $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
3. **Additive Identity:** There exists $\mathbf{0} \in V$ such that $\mathbf{v} + \mathbf{0} = \mathbf{v}$.
4. **Additive Inverse:** For every $\mathbf{v} \in V$, there exists $-\mathbf{v} \in V$ such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$.
5. **Distributivity I:** $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$.
6. **Distributivity II:** $(c + d)\mathbf{v} = c\mathbf{v} + d\mathbf{v}$.
7. **Scalar Associativity:** $c(d\mathbf{v}) = (cd)\mathbf{v}$.
8. **Scalar Identity:** $1 \cdot \mathbf{v} = \mathbf{v}$.

Video Resource: IITM BS Lecture: Introduction to Vector Spaces



Official lecture introducing vector space axioms and canonical examples ($\mathbb{R}^n$, matrix spaces $\mathbb{R}^{m \times n}$, polynomial spaces $\mathbb{P}_n$).

3.2 Vector Subspaces

A subset $W \subseteq V$ of a vector space V is a **Subspace** if W is itself a valid vector space under the same operations.

Theorem / Rule: Subspace Test

A non-empty subset $W \subseteq V$ is a subspace of V if and only if it satisfies **3 closure properties**:

1. **Zero Vector:** $\mathbf{0} \in W$.
2. **Closure under Addition:** If $\mathbf{u}, \mathbf{v} \in W$, then $\mathbf{u} + \mathbf{v} \in W$.
3. **Closure under Scalar Multiplication:** If $\mathbf{v} \in W$ and $c \in \mathbb{R}$, then $c\mathbf{v} \in W$.

Video Resource: IITM BS Lecture: Properties of Vector Spaces & Subspaces 

Official lecture proving subspace criteria and geometric examples (lines/planes through the origin in $\mathbb{R}^3$).

3.3 Linear Combinations, Span, and Linear Independence

Definition: Linear Span and Independence

- **Linear Combination:** $\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k$.
- **Span:** $\text{span}(\{\mathbf{v}_1, \dots, \mathbf{v}_k\})$ is the set of all possible linear combinations of the vectors.
- **Linear Independence:** A set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is **Linearly Independent** if the equation:

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$$

has **ONLY the trivial solution** $c_1 = c_2 = \cdots = c_k = 0$.

If a non-trivial solution exists (at least one $c_i \neq 0$), the vectors are **Linearly Dependent**.

Data Science Relevance: In Machine Learning, **multicollinearity** occurs when input feature vectors are linearly dependent (or nearly dependent). This causes severe instability in linear models and matrix inversion $(X^T X)^{-1}$.

Video Resource: IITM BS Lecture: Linear Dependence 

Official lecture defining linear dependence and redundant vector identification.

Video Resource: IITM BS Lecture: Linear Independence (Parts 1 & 2) 

Official lectures testing linear independence using matrix rank and determinant of coefficient matrices.

3.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Subspace Test:** Determine whether $W = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \mid 2x - 3y = 0 \right\}$ is a valid subspace of $\mathbb{R}^2$. Prove using the 3 subspace criteria.
2. **Non-Subspace Counterexample:** Consider $W = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \mid x + y = 1 \right\}$. Show why W fails to be a vector subspace of $\mathbb{R}^2$.
3. **Testing Linear Independence (3×3):** Test whether the set of vectors:

$$\left\{ \mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 4 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 5 \\ 6 \\ 0 \end{bmatrix} \right\}$$

is linearly independent in $\mathbb{R}^3$ by computing the determinant of the matrix formed by placing them as columns.

4. **Linear Dependence & Redundancy:** Find the scalar value k such that the vectors $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 3 \\ k \end{bmatrix}$ are linearly dependent.
5. **Data Science Application (Feature Multicollinearity):** A data scientist collects three features for housing price prediction: x_1 = size in sq ft, x_2 = size in sq meters, x_3 = number of bedrooms. Since $1 \text{ sq ft} = 0.092903 \text{ sq meters}$, show mathematically that $\{x_1, x_2, x_3\}$ is a linearly dependent set of feature vectors, and explain why one must be removed before regression fitting.

Chapter 4

Basis, Dimension & Matrix Rank

A **Basis** is a minimal generating set of linearly independent vectors that spans a vector space. The number of vectors in any basis defines the unique **Dimension** of the vector space, while **Matrix Rank** counts the dimension of its row and column spaces.

4.1 Basis for a Vector Space

Definition: Basis Definition

A set of vectors $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} \subset V$ is a **Basis** for a vector space V if it satisfies two conditions:

1. $\mathcal{B}$ is **Linearly Independent**.
2. $\mathcal{B}$ **Spans** V , i.e., $\text{span}(\mathcal{B}) = V$.

Every vector $\mathbf{x} \in V$ can be expressed as a **unique** linear combination of the basis vectors.

Example: Standard Basis for $\mathbb{R}^n$

The standard basis for $\mathbb{R}^3$ is $\mathcal{E} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$. Any vector $[x, y, z]^T = x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3$.

Video Resource: IITM BS Lecture: What is a Basis for a Vector Space?



Official lecture introducing basis criteria and unique coordinate representations.

Video Resource: IITM BS Lecture: Finding Bases for Vector Spaces



Official lecture demonstrating methods to extract bases from spanning sets.

4.2 Dimension of a Vector Space

Theorem / Rule: Dimension Theorem

All bases for a given finite-dimensional vector space V contain the **exact same number of vectors**. This invariant number is called the **Dimension** of V , denoted $\dim(V)$.

- $\dim(\mathbb{R}^n) = n$.
- $\dim(\mathbb{R}^{m \times n}) = m \cdot n$.
- The zero subspace $\{0\}$ has dimension 0.

Video Resource: IITM BS Lecture: Rank and Dimension for Vector Spaces 

Official lecture defining dimension, subspace dimension bounds ($\dim(W) \leq \dim(V)$), and basis extensions.

4.3 Matrix Rank via Gaussian Elimination

Key Formula / Algorithm: Column Rank, Row Rank, and Matrix Rank

For any matrix $A \in \mathbb{R}^{m \times n}$:

- **Column Rank:** The dimension of the column space $\text{span}(\text{columns of } A)$.
- **Row Rank:** The dimension of the row space $\text{span}(\text{rows of } A)$.
- **Fundamental Theorem:** $\text{Row Rank}(A) = \text{Column Rank}(A) = \text{rank}(A)$.
- **REF Calculation:** $\text{rank}(A)$ is equal to the **number of non-zero rows (or pivot columns)** in the Row Echelon Form (REF) of A .

Data Science Relevance: In **Principal Component Analysis (PCA)** and low-rank matrix approximations (SVD), matrix rank tells us the effective number of independent latent signals in a dataset!

Video Resource: IITM BS Lecture: Rank and Dimension using Gaussian Elimination 

Official lecture computing matrix rank by row reducing matrices to REF.

4.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Basis Verification:** Check whether $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$ forms a valid basis for $\mathbb{R}^2$. If so, express $\mathbf{v} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$ as a linear combination of $\mathcal{B}$.
2. **Finding a Basis for a Subspace:** Find a basis and the dimension for the subspace of $\mathbb{R}^3$:

$$W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \mid x + 2y - z = 0 \right\}$$

3. **Matrix Rank Calculation:** Compute the rank of the 3×4 matrix A by reducing it to REF:

$$A = \begin{bmatrix} 1 & 2 & 1 & 4 \\ 2 & 4 & 3 & 11 \\ 3 & 6 & 4 & 15 \end{bmatrix}$$

4. **Full Rank vs Rank Deficient:** A dataset matrix $X \in \mathbb{R}^{100 \times 5}$ has 100 samples and 5 features. What is the maximum possible rank of X ? If $\text{rank}(X) = 3$, what does this reveal about the features?
5. **Data Science Application (Collinear Feature Removal):** Given 3 feature vectors:

$$\mathbf{f}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{f}_2 = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}, \quad \mathbf{f}_3 = \begin{bmatrix} 0 \\ 1 \\ 5 \end{bmatrix}$$

Find the rank of the matrix $F = [\mathbf{f}_1 \ \mathbf{f}_2 \ \mathbf{f}_3]$, extract a minimal basis of independent feature vectors, and identify which feature is redundant.

Chapter 5

Null Space, Nullity & Linear Mappings

Matrices are not merely static arrays of numbers—they act as dynamic geometric transformations between vector spaces. In this chapter, we explore the **Null Space** of a matrix and formalize **Linear Mappings** (Linear Transformations).

5.1 Null Space and Nullity

Definition: Null Space ($\text{Null}(A)$ or $\ker(A)$)

For a matrix $A \in \mathbb{R}^{m \times n}$, the **Null Space** (or Kernel) of A is the set of all input vectors $\mathbf{x} \in \mathbb{R}^n$ that map to the zero vector $\mathbf{0} \in \mathbb{R}^m$:

$$\text{Null}(A) = \left\{ \mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0} \right\}$$

The dimension of the null space is called the **Nullity** of A :

$$\text{nullity}(A) = \dim(\text{Null}(A))$$

Theorem / Rule: Null Space is a Subspace

The null space $\text{Null}(A)$ is always a valid vector subspace of the domain $\mathbb{R}^n$. To find a basis for $\text{Null}(A)$, reduce $[A \mid \mathbf{0}]$ to REF/RREF, identify free variables, and solve for the basic variables in terms of the free variables.

Video Resource: IITM BS Lecture: The Null Space of a Matrix (Parts 1 & 2) 

Official lectures defining null space, solving homogenous systems $A\mathbf{x} = \mathbf{0}$, and extracting a basis for $\text{Null}(A)$.

5.2 Linear Mappings and Linear Transformations

A function $T : V \rightarrow W$ between two vector spaces V and W is a **Linear Mapping** (or Linear Transformation) if it preserves vector addition and scalar multiplication.

Definition: Linear Transformation Axioms

A mapping $T : V \rightarrow W$ is **Linear** if for all $\mathbf{u}, \mathbf{v} \in V$ and all scalars $c \in \mathbb{R}$:

1. **Additivity:** $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$.
2. **Homogeneity (Scalar Scaling):** $T(c\mathbf{v}) = cT(\mathbf{v})$.

Together, this implies $T(c_1\mathbf{u} + c_2\mathbf{v}) = c_1T(\mathbf{u}) + c_2T(\mathbf{v})$. Crucial consequence: $T(\mathbf{0}) = \mathbf{0}$ for any linear transformation.

Data Science Relevance: Linear layers in Deep Learning ($\mathbf{y} = W\mathbf{x}$), dimensionality reduction mappings, and continuous linear projections in SVMs are all formal linear transformations!

Video Resource: IITM BS Lecture: What is a Linear Mapping? (Parts 1 & 2) 

Official lectures defining linear mappings, testing additivity and scalar homogeneity.

Video Resource: IITM BS Lecture: What is a Linear Transformation? 

Official lecture demonstrating matrix-induced transformations $T(\mathbf{x}) = A\mathbf{x}$ and geometric transformations (rotations, reflections, shears).

5.3 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Null Space Calculation:** Find a basis for the null space $\text{Null}(A)$ and determine the nullity $\text{nullity}(A)$ for:

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & 5 \end{bmatrix}$$

2. **Testing Linearity of Mappings:** Determine whether each function $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation:

$$(a) \quad T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} 2x - y \\ x + 3y \end{bmatrix}$$

$$(b) \quad T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} x^2 \\ y \end{bmatrix}$$

3. **Matrix-Induced Transformation:** Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be defined by $T(\mathbf{x}) = A\mathbf{x}$, where:

$$A = \begin{bmatrix} 1 & 0 \\ -2 & 3 \\ 5 & 4 \end{bmatrix}$$

Compute $T \left(\begin{bmatrix} 3 \\ -1 \end{bmatrix} \right)$.

4. **Homogenous System Solutions:** A matrix $A \in \mathbb{R}^{3 \times 5}$ has $\text{rank}(A) = 3$. What is the dimension of the domain $\mathbb{R}^n$? What is the nullity $\text{nullity}(A)$? Are there non-trivial solutions to $A\mathbf{x} = \mathbf{0}$?
5. **Data Science Application (Lossless Linear Compression Null Space):** A linear projection layer $W \in \mathbb{R}^{10 \times 50}$ compresses a 50-dimensional input vector $\mathbf{x} \in \mathbb{R}^{50}$ into a 10-dimensional representation $\mathbf{y} = W\mathbf{x}$. What is the minimum possible dimension of information lost in the null space $\text{Null}(W)$?

Chapter 6

Matrix Representation, Kernel, Image & Rank-Nullity

Every linear transformation between finite-dimensional vector spaces can be uniquely represented as a matrix multiplication with respect to chosen ordered bases. Furthermore, the fundamental relationship between domain dimension, image dimension, and kernel dimension is governed by the celebrated **Rank-Nullity Theorem**.

6.1 Matrix Representation of Linear Transformations

Let $T : V \rightarrow W$ be a linear transformation, where $\mathcal{B}_V = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is an ordered basis for domain V , and $\mathcal{B}_W = \{\mathbf{w}_1, \dots, \mathbf{w}_m\}$ is an ordered basis for codomain W .

Definition: Matrix of Transformation $[T]_{\mathcal{B}_V, \mathcal{B}_W}$

The $m \times n$ matrix representing T with respect to bases $\mathcal{B}_V$ and $\mathcal{B}_W$ is constructed by taking the coordinate vectors of $T(\mathbf{v}_j)$ relative to $\mathcal{B}_W$ as its j -th column:

$$A = [T]_{\mathcal{B}_V, \mathcal{B}_W} = \begin{bmatrix} [T(\mathbf{v}_1)]_{\mathcal{B}_W} & [T(\mathbf{v}_2)]_{\mathcal{B}_W} & \dots & [T(\mathbf{v}_n)]_{\mathcal{B}_W} \end{bmatrix} \in \mathbb{R}^{m \times n}$$

Then for any $\mathbf{x} \in V$, coordinate transformation is matrix-vector multiplication: $[T(\mathbf{x})]_{\mathcal{B}_W} = A[\mathbf{x}]_{\mathcal{B}_V}$.

Video Resource: IITM BS Lecture: Linear Transformations, Ordered Bases & Matrices



Official lecture constructing transformation matrices relative to non-standard ordered bases.

6.2 Kernel and Image of Linear Transformations

Definition: Kernel ($\ker(T)$) and Image ($\text{im}(T)$)

- **Kernel (Null Space):** The set of domain vectors mapping to zero:

$$\ker(T) = \left\{ \mathbf{v} \in V \mid T(\mathbf{v}) = \mathbf{0}_W \right\} \subseteq V$$

- **Image (Range / Column Space):** The set of all output vectors in codomain W :

$$\text{im}(T) = \left\{ \mathbf{w} \in W \mid \exists \mathbf{v} \in V \text{ such that } T(\mathbf{v}) = \mathbf{w} \right\} \subseteq W$$

Video Resource: IITM BS Lecture: Image & Kernel of Linear Transformations



Official lecture defining image and kernel subspaces and their geometric interpretations.

Video Resource: IITM BS Lecture: Finding Bases for Kernel and Image



Official lecture computing bases for $\ker(T)$ and $\text{im}(T)$ using REF pivot columns.

6.3 The Rank-Nullity Theorem

Theorem / Rule: The Fundamental Rank-Nullity Theorem

For any linear transformation $T : V \rightarrow W$ on a finite-dimensional domain V :

$$\dim(\ker(T)) + \dim(\text{im}(T)) = \dim(V)$$

In matrix terms for $A \in \mathbb{R}^{m \times n}$:

$$\text{nullity}(A) + \text{rank}(A) = n \quad (\text{number of columns / input features})$$

Data Science Relevance: The Rank-Nullity Theorem guarantees conservation of dimension when passing data through linear autoencoders and neural network linear layers.

6.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Matrix Representation Construction:** Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} 3x + y \\ 2x - 4y \end{bmatrix}$. Find the matrix representation $A = [T]_{\mathcal{E}}$ relative to the standard basis $\mathcal{E}$.
2. **Non-Standard Basis Matrix Representation:** Find the matrix $[T]_{\mathcal{B}}$ for the linear operator $T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} x + y \\ 2x - y \end{bmatrix}$ using the ordered basis $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$.
3. **Bases for Kernel and Image:** Given linear transformation matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$:
 - (a) Find a basis for $\text{im}(A)$ (Column Space).
 - (b) Find a basis for $\text{ker}(A)$ (Null Space).
4. **Rank-Nullity Verification:** Verify the Rank-Nullity Theorem for the matrix A in Exercise 3 by showing $\text{nullity}(A) + \text{rank}(A) = 3$.
5. **Data Science Application (Linear Autoencoder Information Loss):** A linear autoencoder compresses 100-dimensional audio feature vectors $\mathbf{x} \in \mathbb{R}^{100}$ into a bottleneck feature space of dimension 20 using matrix $W \in \mathbb{R}^{20 \times 100}$. If $\text{rank}(W) = 18$, calculate:
 - (a) The dimension of the compressed code space ($\dim(\text{im}(W))$).
 - (b) The dimension of unrecoverable lost features ($\dim(\text{ker}(W))$).

Chapter 7

Matrix Similarity, Affine Spaces, Norms & Inner Products

Geometric intuition in high-dimensional space requires measuring lengths, distances, and angles between vectors. In this chapter, we explore **Matrix Similarity** (change of basis), **Affine Spaces**, and formalize geometric measurement using **Norms** and **Inner Products**.

7.1 Matrix Similarity and Change of Basis

Two square matrices $A, B \in \mathbb{R}^{n \times n}$ are **Similar** if they represent the exact same linear operator $T : V \rightarrow V$ under two different choices of basis.

Theorem / Rule: Matrix Similarity

Matrix B is **similar** to A (written $B \sim A$) if there exists an invertible change-of-basis matrix $P \in \mathbb{R}^{n \times n}$ such that:

$$B = P^{-1}AP$$

Invariants under Similarity: Similar matrices share the exact same:

1. Determinant: $\det(B) = \det(A)$.
2. Trace: $\text{tr}(B) = \text{tr}(A)$.
3. Rank: $\text{rank}(B) = \text{rank}(A)$.
4. Eigenvalues and characteristic polynomial.

Video Resource: IITM BS Lecture: Equivalence & Similarity of Matrices 

Official lecture explaining change-of-basis transformations and similarity invariants.

7.2 Affine Subspaces and Mappings

An **Affine Subspace** (or Flat) is a translated vector subspace: $\mathcal{A} = \mathbf{x}_0 + W$, where $\mathbf{x}_0 \in V$ is a fixed translation point and $W \subseteq V$ is a vector subspace.

Definition: Affine Subspace vs Vector Subspace

Unlike vector subspaces, an affine subspace $\mathbf{x}_0 + W$ does **NOT** need to pass through the origin $\mathbf{0}$ (unless $\mathbf{x}_0 \in W$). For example, lines and planes in $\mathbb{R}^3$ that do not pass through $(0, 0, 0)$ are affine subspaces.

Video Resource: IITM BS Lecture: Affine Subspaces & Mappings 

Official lecture contrasting vector subspaces with shifted affine spaces and hyperplanes.

7.3 Inner Products and Norms

An **Inner Product** $\langle \mathbf{u}, \mathbf{v} \rangle$ generalizes the dot product, providing a rigorous concept of length and angle for any vector space.

Definition: Inner Product Space Axioms

An **Inner Product** $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ satisfies:

1. **Symmetry:** $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$.
2. **Linearity in first argument:** $\langle c\mathbf{u} + d\mathbf{v}, \mathbf{w} \rangle = c\langle \mathbf{u}, \mathbf{w} \rangle + d\langle \mathbf{v}, \mathbf{w} \rangle$.
3. **Positive Definiteness:** $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, and $\langle \mathbf{v}, \mathbf{v} \rangle = 0 \iff \mathbf{v} = \mathbf{0}$.

Every inner product defines an **Induced Norm**: $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$.

Key Formula / Algorithm: Cauchy-Schwarz Inequality & Angle

For any inner product space:

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

The angle θ between non-zero vectors $\mathbf{u}, \mathbf{v}$ is defined by:

$$\cos(\theta) = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

Data Science Relevance: L_1 norm (Manhattan distance, Lasso regularization), L_2 norm (Euclidean distance, Ridge regularization), and Cosine Similarity are core tools in clustering (K-Means), text retrieval, and model loss formulation!

Video Resource: IITM BS Lecture: Lengths, Angles & Inner Products 

Official lecture deriving lengths, unit vectors $\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$, and Cauchy-Schwarz inequality.

Video Resource: IITM BS Lecture: Inner Products and Norms on a Vector Space 

Official lecture introducing weighted inner products $\langle \mathbf{u}, \mathbf{v} \rangle_Q = \mathbf{u}^T Q \mathbf{v}$ with positive-definite Q .

7.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Matrix Similarity Check:** Given $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 0 \\ 1 & 1 \end{bmatrix}$:
 - (a) Compute $\det(A)$ and $\det(B)$.
 - (b) Compute $\text{tr}(A)$ and $\text{tr}(B)$.
 - (c) Are A and B similar matrices?
2. **Weighted Inner Product:** Define a weighted inner product on $\mathbb{R}^2$ by $\langle \mathbf{u}, \mathbf{v} \rangle_Q = 2u_1v_1 + 5u_2v_2$.
 - (a) Compute $\left\langle \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \end{bmatrix} \right\rangle_Q$.
 - (b) Compute the induced norm $\left\| \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\|_Q$.

3. **Cosine Similarity:** In natural language processing, word embeddings for two documents are:

$$\mathbf{d}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{d}_2 = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Compute the Cosine Similarity $\cos(\theta) = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{\|\mathbf{d}_1\| \|\mathbf{d}_2\|}$.

4. **Cauchy-Schwarz Verification:** For $\mathbf{u} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ in standard $\mathbb{R}^2$, verify that $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$.
5. **Data Science Application (L_1 vs L_2 Norm Regularization):** A weights vector in a sparse ML model is $\mathbf{w} = \begin{bmatrix} 3 \\ -4 \\ 0 \\ 12 \end{bmatrix}$.

- (a) Calculate the L_1 norm $\|\mathbf{w}\|_1 = \sum |w_i|$ (Lasso penalty).
- (b) Calculate the L_2 norm $\|\mathbf{w}\|_2 = \sqrt{\sum w_i^2}$ (Ridge penalty).

Chapter 8

Orthogonality, Gram-Schmidt Process & Projections

Orthogonality is the mathematical formalization of perpendicularity. Orthogonal vectors provide independent, uncorrelated coordinates. In this chapter, we master **Orthonormal Bases**, **Orthogonal Projections**, and the **Gram-Schmidt Process**.

8.1 Orthogonality and Orthonormal Bases

Definition: Orthogonality & Orthonormal Basis

- Two vectors $\mathbf{u}, \mathbf{v}$ are **Orthogonal** if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ (written $\mathbf{u} \perp \mathbf{v}$).
- A set of vectors $\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$ is an **Orthonormal Basis (ONB)** if:

$$\langle \mathbf{q}_i, \mathbf{q}_j \rangle = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \quad (\text{Unit Norm: } \|\mathbf{q}_i\| = 1) \\ 0 & \text{if } i \neq j \quad (\text{Mutually Orthogonal}) \end{cases}$$

Theorem / Rule: Coordinate Computation with ONB

If $\mathcal{B} = \{\mathbf{q}_1, \dots, \mathbf{q}_n\}$ is an ONB for V , the coordinates of any $\mathbf{v} \in V$ are obtained simply by taking inner products without solving linear systems!

$$\mathbf{v} = \sum_{i=1}^n \langle \mathbf{v}, \mathbf{q}_i \rangle \mathbf{q}_i = \langle \mathbf{v}, \mathbf{q}_1 \rangle \mathbf{q}_1 + \dots + \langle \mathbf{v}, \mathbf{q}_n \rangle \mathbf{q}_n$$

Video Resource: IITM BS Lecture: Orthogonality & Linear Independence 

Official lecture proving that any set of non-zero mutually orthogonal vectors is automatically linearly independent.

Video Resource: IITM BS Lecture: What is an Orthonormal Basis? 

Official lecture defining ONB and showing simple coordinate extraction via inner products.

8.2 Orthogonal Projections

The **Orthogonal Projection** of a vector $\mathbf{v}$ onto a subspace W finds the unique point $\mathbf{p} = \text{proj}_W(\mathbf{v}) \in W$ that minimizes the distance $\|\mathbf{v} - \mathbf{p}\|$.

Key Formula / Algorithm: Projection onto Subspace W with ONB $\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$

$$\text{proj}_W(\mathbf{v}) = \sum_{i=1}^k \langle \mathbf{v}, \mathbf{q}_i \rangle \mathbf{q}_i$$

The error vector $\mathbf{e} = \mathbf{v} - \text{proj}_W(\mathbf{v})$ is strictly orthogonal to W ($\mathbf{e} \perp W$).

Video Resource: IITM BS Lecture: Projections Using Inner Products 

Official lecture deriving projection formulas, best approximation theorem, and error orthogonality.

8.3 The Gram-Schmidt Orthogonalization Process

The **Gram-Schmidt Process** converts an arbitrary linearly independent basis $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ into an orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_k\}$.

Key Formula / Algorithm: Gram-Schmidt Algorithm

1. **First Vector:** $\mathbf{u}_1 = \mathbf{v}_1$, $\mathbf{q}_1 = \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|}$ 2. **Second Vector:** Subtract projection onto $\mathbf{q}_1$:

$$\mathbf{u}_2 = \mathbf{v}_2 - \langle \mathbf{v}_2, \mathbf{q}_1 \rangle \mathbf{q}_1, \quad \mathbf{q}_2 = \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|}$$

3. **k -th Vector:** Subtract projections onto all previous $\mathbf{q}_j$:

$$\mathbf{u}_k = \mathbf{v}_k - \sum_{j=1}^{k-1} \langle \mathbf{v}_k, \mathbf{q}_j \rangle \mathbf{q}_j, \quad \mathbf{q}_k = \frac{\mathbf{u}_k}{\|\mathbf{u}_k\|}$$

Data Science Relevance: The Gram-Schmidt process is the foundation of **QR Factorization** ($A = QR$) used in stable Ordinary Least Squares regression and numerical eigenvalue algorithms!

Video Resource: IITM BS Lecture: The Gram-Schmidt Process 

Official lecture executing Gram-Schmidt step-by-step to construct orthonormal bases.

Video Resource: IITM BS Lecture: Orthogonal Transformations & Rotations 

Official lecture introducing orthogonal matrices $Q^T Q = I$ (preserves lengths and angles) and rotation matrices.

8.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Projection onto a Vector:** Compute the orthogonal projection of vector $\mathbf{v} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ onto the line spanned by $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.
2. **Gram-Schmidt Process (2×2):** Use the Gram-Schmidt process to convert the basis $\mathcal{B} = \left\{ \mathbf{v}_1 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$ into an orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2\}$.
3. **Gram-Schmidt Process (3×3):** Apply Gram-Schmidt to orthogonalize $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.
4. **Orthogonal Matrix Properties:** Let $Q = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ be a 2D rotation matrix.
 - (a) Compute $Q^T Q$.
 - (b) Show that $\|Q\mathbf{x}\| = \|\mathbf{x}\|$ for any $\mathbf{x} \in \mathbb{R}^2$ (preserves norm).
5. **Data Science Application (Least Squares via Projection):** In linear regression, the best fit prediction vector $\hat{\mathbf{y}} = X\mathbf{w}$ is the orthogonal projection of target vector $\mathbf{y}$ onto the column space of feature matrix X :

$$\hat{\mathbf{y}} = P\mathbf{y}, \quad \text{where } P = X(X^T X)^{-1} X^T$$

Show that the projection matrix P is symmetric ($P^T = P$) and idempotent ($P^2 = P$).

Chapter 9

Multivariable Functions, Partial Derivatives & Gradient

Transitioning from 1D single-variable calculus to **Multivariable Calculus** allows us to model functions with multiple input features: $f(\mathbf{x}) = f(x_1, x_2, \dots, x_n)$. In this chapter, we master **Partial Derivatives**, **Directional Derivatives**, and the **Gradient Vector** ∇f .

9.1 Multivariable Functions and Visualizations

A scalar-valued multivariable function maps a feature vector in $\mathbb{R}^n$ to a real number output:

$$f : \mathbb{R}^n \rightarrow \mathbb{R}$$

Definition: Level Sets and Contour Curves

For $f(x, y)$, a **Level Curve** (or Contour) is the set of all points (x, y) where the function takes a constant value c :

$$\{(x, y) \in \mathbb{R}^2 \mid f(x, y) = c\}$$

Video Resource: IITM BS Lecture: Multivariable Functions & Visualization 

Official lecture visualizing 3D surface plots $z = f(x, y)$ and 2D contour lines.

9.2 Partial Derivatives

The **Partial Derivative** of $f(x_1, x_2, \dots, x_n)$ with respect to x_i measures the rate of change of f along the x_i coordinate axis, holding all other variables constant.

Theorem / Rule: Partial Derivative Calculation

To compute $\frac{\partial f}{\partial x_i}$ (or f_{x_i}), treat all variables except x_i as constants and differentiate using standard 1D calculus rules.

$$\frac{\partial f}{\partial x_1} = \lim_{h \rightarrow 0} \frac{f(x_1 + h, x_2, \dots, x_n) - f(x_1, x_2, \dots, x_n)}{h}$$

Example: Computing Partial Derivatives

For $f(x, y) = x^3y^2 + 5x^2 - 4y$:

$$\frac{\partial f}{\partial x} = 3x^2y^2 + 10x$$

$$\frac{\partial f}{\partial y} = 2x^3y - 4$$

Video Resource: IITM BS Lecture: Partial Derivatives



Official lecture defining partial derivatives geometrically as tangent slopes along coordinate grid slices.

9.3 The Gradient Vector and Directional Derivatives

The **Gradient** $\nabla f(\mathbf{x})$ packs all first-order partial derivatives into a single column vector.

Definition: The Gradient Vector (∇f)

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the **Gradient Vector** is:

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix} \in \mathbb{R}^n$$

Key Formula / Algorithm: Directional Derivative via Gradient

The **Directional Derivative** $D_{\mathbf{u}}f(\mathbf{x})$ measures the rate of change of f in the direction of a **unit vector** $\mathbf{u}$ ($\|\mathbf{u}\| = 1$):

$$D_{\mathbf{u}}f(\mathbf{x}) = \nabla f(\mathbf{x}) \cdot \mathbf{u} = \|\nabla f(\mathbf{x})\| \cos \theta$$

Data Science Relevance: The gradient $\nabla f(\mathbf{x})$ points in the direction of **steepest increase** of the function, while $-\nabla f(\mathbf{x})$ points in the direction of **steepest descent**—the core mechanism behind updating neural network weights during Gradient Descent training!

Video Resource: IITM BS Lecture: Directional Derivatives



Official lecture defining rate of change along arbitrary unit vector directions.

Video Resource: IITM BS Lecture: Directional Derivatives in Terms of Gradient



Official lecture proving $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$ and geometric properties of gradients.

9.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **First-Order Partial Derivatives:** Compute $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for:

$$f(x, y) = x^2 y^3 + e^{2x} \sin(y)$$

2. **Gradient Vector Evaluation:** Compute the gradient vector $\nabla f(1, 2)$ for:

$$f(x, y) = 4x^2 y - 3y^2 + 5x$$

3. **Directional Derivative:** Calculate the directional derivative $D_{\mathbf{u}}f(2, 1)$ for $f(x, y) = x^2 + 3xy + y^2$ in the direction of vector $\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. (Remember to normalize $\mathbf{v}$ into a unit vector $\mathbf{u}$).
4. **Steepest Ascent Direction:** At the point $(1, -1)$, in which unit vector direction does $f(x, y) = x^3 y - 2xy^2$ increase most rapidly? What is the maximum rate of increase?
5. **Data Science Application (MSE Loss Gradient):** In linear regression with 1 sample (x_i, y_i) , the Mean Squared Error loss as a function of slope m and bias b is:

$$L(m, b) = (mx_i + b - y_i)^2$$

Compute the gradient vector $\nabla L(m, b) = \begin{bmatrix} \frac{\partial L}{\partial m} \\ \frac{\partial L}{\partial b} \end{bmatrix}$.

Chapter 10

Tangent Planes, Steepest Descent & Critical Points

Optimization is the heart of Machine Learning. Training a model means finding parameter values that minimize a loss function $L(\mathbf{w})$. In this chapter, we master **Tangent Planes/Hyperplanes**, the **Gradient Descent Update Rule**, and **Critical Points** ($\nabla f(\mathbf{x}) = 0$).

10.1 Tangent Planes and Hyperplanes

The 1D tangent line generalizes in higher dimensions to a **Tangent Plane** (for $f : \mathbb{R}^2 \rightarrow \mathbb{R}$) or a **Tangent Hyperplane** (for $f : \mathbb{R}^n \rightarrow \mathbb{R}$).

Definition: Tangent Plane Formula

For a surface $z = f(x, y)$, the tangent plane at point (x_0, y_0, z_0) where $z_0 = f(x_0, y_0)$ is given by:

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

In general for $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the linear approximation (tangent hyperplane) at $\mathbf{x}_0$ is:

$$L(\mathbf{x}) = f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^T (\mathbf{x} - \mathbf{x}_0)$$

Video Resource: IITM BS Lecture: Tangents for Scalar-Valued Functions 

Official lecture deriving tangent plane equations and linear approximations.

Video Resource: IITM BS Lecture: Finding the Tangent Hyperplane 

Official lecture extending tangent planes to n -dimensional hyperplanes $\nabla f(\mathbf{x}_0)^T (\mathbf{x} - \mathbf{x}_0) = 0$.

10.2 Steepest Descent and Gradient Descent Algorithm

Theorem / Rule: Direction of Steepest Descent

- **Steepest Ascent:** The unit vector direction along which $f(\mathbf{x})$ increases fastest is $\frac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|}$.
- **Steepest Descent:** The unit vector direction along which $f(\mathbf{x})$ decreases fastest is $-\frac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|}$.

Key Formula / Algorithm: Gradient Descent Update Rule

To minimize a cost function $L(\mathbf{w})$, start at an initial point $\mathbf{w}^{(0)}$ and iteratively update:

$$\mathbf{w}^{(k+1)} = \mathbf{w}^{(k)} - \alpha \nabla L(\mathbf{w}^{(k)})$$

where $\alpha > 0$ is the **Learning Rate** (step size).

Data Science Relevance: Gradient Descent (and its variants like Adam, RMSprop, SGD) is used to train almost all modern Machine Learning models, from Logistic Regression to 70-Billion parameter Large Language Models!

Video Resource: IITM BS Lecture: The Direction of Steepest Ascent / Descent



Official lecture proving why $-\nabla f$ is the optimal search direction for loss minimization.

10.3 Critical Points

Definition: Critical Points

An interior point $\mathbf{x}_0$ in the domain of f is a **Critical Point** (or Stationary Point) if:

$$\nabla f(\mathbf{x}_0) = \mathbf{0} \iff \frac{\partial f}{\partial x_1}(\mathbf{x}_0) = 0, \frac{\partial f}{\partial x_2}(\mathbf{x}_0) = 0, \dots, \frac{\partial f}{\partial x_n}(\mathbf{x}_0) = 0$$

Critical points can be **Local Minima**, **Local Maxima**, or **Saddle Points**.

Video Resource: IITM BS Lecture: Critical Points for Multivariable Functions



Official lecture finding critical points by setting all first partial derivatives to zero simultaneously.

10.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Tangent Plane Equation:** Find the equation of the tangent plane to the surface $z = x^2 + 3y^2 - 4x$ at the point $(x_0, y_0) = (2, 1)$.
2. **Gradient Descent Iteration Step:** A 2D loss function is $L(w_1, w_2) = w_1^2 + 4w_2^2$.
 - (a) Compute the gradient $\nabla L(w_1, w_2)$.
 - (b) Starting from $\mathbf{w}^{(0)} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ with learning rate $\alpha = 0.1$, compute the next weight vector $\mathbf{w}^{(1)}$.

3. **Finding Critical Points (2×2):** Find all critical points of the function:

$$f(x, y) = x^3 + y^3 - 3xy$$

by setting $f_x = 0$ and $f_y = 0$.

4. **Linear Approximation:** Use the tangent plane linear approximation of $f(x, y) = \sqrt{x^2 + y^2}$ at $(3, 4)$ to estimate $f(3.06, 3.92)$.
5. **Data Science Application (Least Squares Gradient Descent Update):** For a simple regression model with loss $L(w) = (2w - 6)^2$, start at $w^{(0)} = 0$ with learning rate $\alpha = 0.05$. Perform 2 steps of Gradient Descent to find $w^{(1)}$ and $w^{(2)}$. Show that w moves towards the true minimum $w^* = 3$.

Chapter 11

Hessian Matrix, Multivariable Extrema & Differentiability

To determine whether a critical point $\mathbf{x}_0$ is a local minimum, local maximum, or saddle point, we analyze second-order partial derivatives. In this chapter, we master the **Hessian Matrix**, the **Second Derivative Test**, and formal **Multivariable Differentiability**.

11.1 Second-Order Partial Derivatives and Clairaut's Theorem

For $f(x, y)$, there are four second-order partial derivatives:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \quad f_{yy} = \frac{\partial^2 f}{\partial y^2}, \quad f_{xy} = \frac{\partial^2 f}{\partial y \partial x}, \quad f_{yx} = \frac{\partial^2 f}{\partial x \partial y}$$

Theorem / Rule: Clairaut's Theorem (Symmetry of Mixed Partial)

If f_{xy} and f_{yx} are continuous on an open set containing $\mathbf{x}_0$, then the mixed partials are equal:

$$f_{xy} = f_{yx}$$

11.2 The Hessian Matrix

Definition: The Hessian Matrix $H(f)$

The **Hessian Matrix** $H(f)(\mathbf{x})$ is the symmetric matrix of second-order partial derivatives:

$$H(f)(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix} \in \mathbb{R}^{n \times n}$$

Video Resource: IITM BS Lecture: Higher Order Partial Derivatives & Hessian Matrix



Official lecture constructing Hessian matrices for 2D and 3D functions.

11.3 Second Derivative Test for Local Extrema

Theorem / Rule: Second Derivative Test for $f(x, y)$

Let (x_0, y_0) be a critical point ($\nabla f(x_0, y_0) = \mathbf{0}$). Define the Hessian determinant D :

$$D = \det(H(f)(x_0, y_0)) = f_{xx}f_{yy} - (f_{xy})^2$$

1. If $D > 0$ and $f_{xx} > 0$ (Hessian is Positive Definite): (x_0, y_0) is a **Local Minimum**.
2. If $D > 0$ and $f_{xx} < 0$ (Hessian is Negative Definite): (x_0, y_0) is a **Local Maximum**.
3. If $D < 0$ (Hessian is Indefinite): (x_0, y_0) is a **Saddle Point**.
4. If $D = 0$: Test is **Inconclusive**.

Data Science Relevance: In Optimization and Deep Learning, Newton-Raphson optimization uses the inverse Hessian $H^{-1}\nabla f$ for second-order gradient updates. Convex loss functions (where $H(L)$ is positive semi-definite everywhere) guarantee that any local minimum is a global minimum!

Video Resource: IITM BS Lecture: Hessian Matrix & Local Extrema for $f(x, y)$ 

Official lecture classifying 2D critical points using the Hessian discriminant D .

Video Resource: IITM BS Lecture: Hessian Matrix for 3D Functions $f(x, y, z)$ 

Official lecture extending local extrema classification to 3D via leading principal minors of $H(f)$.

11.4 Multivariable Differentiability

Unlike 1D functions, the existence of partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ does **NOT** guarantee that a multivariable function is continuous or differentiable!

Definition: Formal Multivariable Differentiability

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is **Differentiable** at $\mathbf{x}_0$ if there exists a linear mapping (the derivative total transformation J) such that:

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - \nabla f(\mathbf{x}_0)^T \mathbf{h}|}{\|\mathbf{h}\|} = 0$$

Sufficient Condition: If all first partial derivatives $\frac{\partial f}{\partial x_i}$ exist and are **continuous** near $\mathbf{x}_0$, then f is guaranteed to be differentiable at $\mathbf{x}_0$.

Video Resource: IITM BS Lecture: Differentiability for Multivariable Functions 

Official lecture contrasting partial derivative existence with formal multivariable differentiability.

11.5 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Hessian Matrix Computation:** Compute the Hessian matrix $H(f)(x, y)$ for $f(x, y) = x^3 - 3xy^2 + y^4$. Evaluate it at the point $(1, 2)$.
2. **Classifying 2D Critical Points:** Find all critical points of $f(x, y) = x^3 + y^3 - 3xy$ (from Week 10 exercise: $(0, 0)$ and $(1, 1)$), and classify each point as a local minimum, local maximum, or saddle point using the Hessian test.
3. **Saddle Point Identification:** Show that the origin $(0, 0)$ is a saddle point for the hyperbolic paraboloid $f(x, y) = x^2 - y^2$ by analyzing the eigenvalues or determinant of $H(f)(0, 0)$.
4. **Classifying 3D Critical Points:** For $f(x, y, z) = x^2 + 2y^2 + 3z^2 - 4x + 8y - 12z$, find the critical point and construct the 3×3 Hessian matrix to classify it.
5. **Data Science Application (Convexity of Mean Squared Error):** The OLS Linear Regression loss function for 1 feature w is $L(w) = \frac{1}{N} \sum_{i=1}^N (wx_i - y_i)^2$.
 - (a) Compute the second derivative $\frac{d^2 L}{dw^2}$.
 - (b) Show that $\frac{d^2 L}{dw^2} \geq 0$ for all w , proving that the loss surface is strictly convex and has a unique global minimum.

Chapter 12

Eigenvalues, Eigenvectors, Diagonalization & SVD

Eigenvalues and Eigenvectors form the ultimate cornerstone of modern linear algebra and data science. In this capstone chapter, we master **Eigen-decomposition**, **Matrix Diagonalization**, and **Singular Value Decomposition (SVD)**.

12.1 Eigenvalues and Eigenvectors

When a matrix $A \in \mathbb{R}^{n \times n}$ acts on a vector $\mathbf{v}$, it typically changes both the magnitude and direction of $\mathbf{v}$. However, certain non-zero vectors maintain their original direction and are merely scaled by a factor λ .

Definition: Eigenvalue Problem

A non-zero vector $\mathbf{v} \neq \mathbf{0}$ is an **Eigenvector** of A with corresponding **Eigenvalue** $\lambda \in \mathbb{R}$ (or $\mathbb{C}$) if:

$$A\mathbf{v} = \lambda\mathbf{v}$$

Equivalently:

$$(A - \lambda I)\mathbf{v} = \mathbf{0}$$

Theorem / Rule: The Characteristic Equation

Since $\mathbf{v} \neq \mathbf{0}$, the matrix $(A - \lambda I)$ must be singular ($\det(A - \lambda I) = 0$). 1. **Characteristic Polynomial:** $p(\lambda) = \det(A - \lambda I) = 0$. Solve for roots $\lambda_1, \lambda_2, \dots, \lambda_n$. 2. **Eigenspace:** For each λ_i , solve $(A - \lambda_i I)\mathbf{v} = \mathbf{0}$ to find the basis of eigenvectors $\text{Null}(A - \lambda_i I)$.

Video Resource: 3Blue1Brown: Essence of Linear Algebra (Eigenvalues & Eigenvectors)



Visual intuition showing eigenvectors as axes of rotation/scaling during linear transformations.

12.2 Matrix Diagonalization

Definition: Diagonalization Theorem

An $n \times n$ matrix A is **Diagonalizable** if it has n linearly independent eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$. Form eigenvector matrix $P = [\mathbf{v}_1 \dots \mathbf{v}_n]$ and diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_n)$. Then:

$$A = PDP^{-1} \iff D = P^{-1}AP$$

Matrix Powers: $A^k = PD^kP^{-1}$, making high-power matrix computations $\mathcal{O}(n)$ instead of $\mathcal{O}(n^k)$!

Theorem / Rule: Spectral Theorem for Symmetric Matrices

If $A = A^T$ is a real symmetric matrix, then: 1. All eigenvalues of A are real numbers. 2. Eigenvectors corresponding to distinct eigenvalues are **mutually orthogonal**. 3. A is **orthogonally diagonalizable**: $A = QDQ^T$, where Q is an orthogonal matrix ($Q^TQ = I$).

12.3 Singular Value Decomposition (SVD)

While Eigen-decomposition requires square matrices $A \in \mathbb{R}^{n \times n}$, **Singular Value Decomposition (SVD)** applies to ANY rectangular matrix $A \in \mathbb{R}^{m \times n}$.

Key Formula / Algorithm: Singular Value Decomposition ($A = U\Sigma V^T$)

Every matrix $A \in \mathbb{R}^{m \times n}$ can be factored into:

$$A = U\Sigma V^T$$

where:

- $U \in \mathbb{R}^{m \times m}$: Left singular vectors (orthonormal eigenvectors of AA^T).
- $\Sigma \in \mathbb{R}^{m \times n}$: Diagonal matrix of singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$ ($\sigma_i = \sqrt{\lambda_i(A^T A)}$).
- $V \in \mathbb{R}^{n \times n}$: Right singular vectors (orthonormal eigenvectors of $A^T A$).

```

1 import numpy as np
2
3 # 1. Eigenvalue Decomposition
4 A = np.array([[4.0, 1.0],
5               [2.0, 3.0]])
6
7 eigenvalues, eigenvectors = np.linalg.eig(A)
8 print("Eigenvalues:", eigenvalues)
9 print("Eigenvectors:\n", eigenvectors)
10
11 # Verify A @ v = lambda * v for the first eigenvector
12 v0 = eigenvectors[:, 0]
13 lam0 = eigenvalues[0]
14 print("A @ v0:      ", A @ v0)
15 print("lambda * v0:", lam0 * v0)

```



```
16
17 # 2. Singular Value Decomposition (SVD)
18 X = np.array([[3.0, 1.0],
19               [1.0, 3.0],
20               [0.0, 2.0]])
21
22 U, S, Vt = np.linalg.svd(X)
23 print("\nLeft Singular Vectors U shape:", U.shape)
24 print("Singular Values S:", S)
25 print("Right Singular Vectors V^T shape:", Vt.shape)
```

Listing 12.1: NumPy Implementation of Eigen-decomposition and SVD

Video Resource: StatQuest: Singular Value Decomposition (SVD) & PCA 

Step-by-step visual explanation of SVD matrix factorization and dimensionality reduction.

Video Resource: IITM BS Lecture: Mathematics for Data Science II Review 

Official comprehensive course summary reviewing Linear Algebra and Multivariable Calculus.

12.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Eigenvalues & Eigenvectors (2×2):** Find the eigenvalues λ_1, λ_2 and corresponding eigenvectors $\mathbf{v}_1, \mathbf{v}_2$ for the matrix:

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

2. **Matrix Diagonalization ($A = PDP^{-1}$):** Using the eigenvalues and eigenvectors from Exercise 1, construct the matrices P and D such that $A = PDP^{-1}$. Compute A^4 using PD^4P^{-1} .
3. **Symmetric Matrix Spectral Decomposition:** Verify that $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ is symmetric. Find its real eigenvalues and construct an orthogonal matrix Q ($Q^T Q = I$) such that $A = QDQ^T$.
4. **Singular Values Calculation:** Given matrix $A = \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}$, compute the matrix $A^T A$, find its eigenvalues λ_i , and determine the singular values σ_1, σ_2 of A .
5. **Comprehensive Course Review Question (PCA via SVD):** A data matrix $X \in \mathbb{R}^{N \times d}$ has covariance matrix $C = \frac{1}{N} X^T X$.
 - (a) Explain why the principal component directions are the eigenvectors of C .
 - (b) If the eigenvalues of C are $\lambda_1 = 80, \lambda_2 = 15, \lambda_3 = 5$, what percentage of total dataset variance is explained by keeping the first 2 principal components?

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Solutions to Practice Exercises

This appendix contains complete, step-by-step mathematical solutions for all practice exercises across Weeks 1 to 12.

Week 1: Vectors, Matrices & Determinants

Detailed Solutions

1. Vector Dot Product & Norms

- (a) Dot Product:

$$\mathbf{u} \cdot \mathbf{v} = (2)(1) + (3)(-2) + (-1)(4) = 2 - 6 - 4 = -8$$

- (b) Norms:

$$\|\mathbf{u}\| = \sqrt{2^2 + 3^2 + (-1)^2} = \sqrt{4 + 9 + 1} = \sqrt{14}$$

$$\|\mathbf{v}\| = \sqrt{1^2 + (-2)^2 + 4^2} = \sqrt{1 + 4 + 16} = \sqrt{21}$$

2. Matrix Multiplication

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 1(0) + 2(2) & 1(-1) + 2(5) \\ 3(0) + 4(2) & 3(-1) + 4(5) \end{bmatrix} = \begin{bmatrix} 4 & 9 \\ 8 & 17 \end{bmatrix}$$

$$BA = \begin{bmatrix} 0 & -1 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 0(1) - 1(3) & 0(2) - 1(4) \\ 2(1) + 5(3) & 2(2) + 5(4) \end{bmatrix} = \begin{bmatrix} -3 & -4 \\ 17 & 24 \end{bmatrix}$$

Since $\begin{bmatrix} 4 & 9 \\ 8 & 17 \end{bmatrix} \neq \begin{bmatrix} -3 & -4 \\ 17 & 24 \end{bmatrix}$, matrix multiplication is non-commutative ($AB \neq BA$).

3. Matrix-Vector System Formulation

$$A = \begin{bmatrix} 2 & 3 & -1 \\ 4 & -1 & 2 \\ -1 & 5 & 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 7 \\ 3 \\ 10 \end{bmatrix}$$

Augmented Matrix $[A \mid \mathbf{b}]$:

$$[A \mid \mathbf{b}] = \left[\begin{array}{ccc|c} 2 & 3 & -1 & 7 \\ 4 & -1 & 2 & 3 \\ -1 & 5 & 3 & 10 \end{array} \right]$$

4. Determinant Calculation Expand M along row 1:

$$\det(M) = 3 \begin{vmatrix} 4 & 5 \\ 2 & 1 \end{vmatrix} - 1 \begin{vmatrix} 0 & 5 \\ 1 & 1 \end{vmatrix} + 2 \begin{vmatrix} 0 & 4 \\ 1 & 2 \end{vmatrix}$$

$$\det(M) = 3(4 - 10) - 1(0 - 5) + 2(0 - 4) = 3(-6) - 1(-5) + 2(-4) = -18 + 5 - 8 = -21$$

Since $\det(M) = -21 \neq 0$, the matrix M is **invertible**.

5. Data Science Application: Covariance Determinant

$$\det(\Sigma) = \begin{vmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{vmatrix} = \sigma_1^2 \sigma_2^2 - \sigma_{12}^2$$

$$\det(\Sigma) = (4)(9) - (3)^2 = 36 - 9 = \mathbf{27}$$

Week 2: Cramer's Rule, Row Echelon Form & Gaussian Elimination

Detailed Solutions

1. Cramer's Rule (2×2)

$$\det(A) = \begin{vmatrix} 3 & 4 \\ 1 & -2 \end{vmatrix} = 3(-2) - 4(1) = -6 - 4 = \mathbf{-10}$$

$$\det(A_1(\mathbf{b})) = \begin{vmatrix} 10 & 4 \\ -3 & -2 \end{vmatrix} = 10(-2) - 4(-3) = -20 + 12 = -8 \implies x_1 = \frac{-8}{-10} = \mathbf{0.8}$$

$$\det(A_2(\mathbf{b})) = \begin{vmatrix} 3 & 10 \\ 1 & -3 \end{vmatrix} = 3(-3) - 10(1) = -9 - 10 = -19 \implies x_2 = \frac{-19}{-10} = \mathbf{1.9}$$

2. Identifying REF

- **(a) REF:** Pivots are at (1, 1) and (2, 2), entry below pivots is 0, zero row is at bottom.
- **(b) REF:** Upper triangular with pivots at (1, 1), (2, 2), (3, 3).
- **(c) Neither:** The pivot in row 2 is in column 1, which is to the left of the pivot in row 1 (column 2).

3. Gaussian Elimination (3×3) Augmented matrix $[A \mid \mathbf{b}]$:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 2 & 3 & 4 & 20 \\ 4 & 3 & 2 & 16 \end{array} \right]$$

1. $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 4R_1$:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & -1 & 2 & 4 \\ 0 & -5 & -2 & -16 \end{array} \right]$$

2. $R_3 \rightarrow R_3 - 5R_2$:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & -1 & 2 & 4 \\ 0 & 0 & -12 & -36 \end{array} \right]$$

3. Back-Substitution:

- Row 3: $-12x_3 = -36 \implies \mathbf{x_3 = 3}$.
- Row 2: $-x_2 + 2(3) = 4 \implies -x_2 + 6 = 4 \implies \mathbf{x_2 = 2}$.
- Row 1: $x_1 + 2(2) + 3 = 8 \implies x_1 + 7 = 8 \implies \mathbf{x_1 = 1}$.

Solution: $(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) = (1, 2, 3)$.

4. Free Variables & Infinite Solutions

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 2 & 4 & 1 & 11 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 0 & 3 & 3 \end{array} \right]$$

- Row 2: $3x_3 = 3 \implies \mathbf{x}_3 = 1$.
- Row 1: $x_1 + 2x_2 - 1 = 4 \implies x_1 = 5 - 2x_2$.

Here x_2 is a **free variable** ($x_2 = t$). General Parametric Solution: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 - 2t \\ t \\ 1 \end{bmatrix}$ for any $t \in \mathbb{R}$.

5. Data Science Application: Least Squares Normal Equations Using Cramer's Rule on $\begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} = \begin{bmatrix} 10 \\ 9 \end{bmatrix}$:

$$\det(A) = 4(3) - 2(2) = 12 - 4 = 8$$

$$w_0 = \frac{\begin{vmatrix} 10 & 2 \\ 9 & 3 \end{vmatrix}}{8} = \frac{30 - 18}{8} = \frac{12}{8} = \mathbf{1.5}$$

$$w_1 = \frac{\begin{vmatrix} 4 & 10 \\ 2 & 9 \end{vmatrix}}{8} = \frac{36 - 20}{8} = \frac{16}{8} = \mathbf{2.0}$$

Parameter Vector: $\mathbf{w} = \begin{bmatrix} 1.5 \\ 2.0 \end{bmatrix}$.

Week 3: Vector Spaces, Subspaces & Linear Independence

Detailed Solutions

1. Subspace Test for $W = \{[x, y]^T \mid 2x - 3y = 0\}$

- 1. Zero Vector:** For $\mathbf{0} = [0, 0]^T$, $2(0) - 3(0) = 0$. $\mathbf{0} \in W$.
- 2. Closure under Addition:** Let $\mathbf{u} = [x_1, y_1]^T, \mathbf{v} = [x_2, y_2]^T \in W$. Then $2x_1 - 3y_1 = 0$ and $2x_2 - 3y_2 = 0$. For $\mathbf{u} + \mathbf{v} = [x_1 + x_2, y_1 + y_2]^T$:

$$2(x_1 + x_2) - 3(y_1 + y_2) = (2x_1 - 3y_1) + (2x_2 - 3y_2) = 0 + 0 = 0 \implies \mathbf{u} + \mathbf{v} \in W$$

- 3. Closure under Scalar Multiplication:** For any scalar $c \in \mathbb{R}$:

$$2(cx_1) - 3(cy_1) = c(2x_1 - 3y_1) = c(0) = 0 \implies c\mathbf{u} \in W$$

Conclusion: W satisfies all 3 criteria and is a **valid subspace of $\mathbb{R}^2$** .

2. Non-Subspace Counterexample For $W = \{[x, y]^T \mid x + y = 1\}$:

- **Zero vector test:** $\mathbf{0} = [0, 0]^T \implies 0 + 0 = 0 \neq 1$. $\mathbf{0} \notin W$.

Failing the zero vector test immediately proves W is **NOT a vector subspace**.

3. Testing Linear Independence (3×3) Form matrix $A = [\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3] = \begin{bmatrix} 1 & 0 & 5 \\ 2 & 1 & 6 \\ 3 & 4 & 0 \end{bmatrix}$. Compute

$\det(A)$ expanding along row 1:

$$\det(A) = 1(1 \cdot 0 - 4 \cdot 6) - 0 + 5(2 \cdot 4 - 3 \cdot 1) = 1(-24) + 5(8 - 3) = -24 + 25 = \mathbf{1}$$

Since $\det(A) = 1 \neq 0$, the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is **Linearly Independent**.

4. Linear Dependence Condition Vectors are linearly dependent iff $\det([\mathbf{u} \ \mathbf{v}]) = 0$:

$$\begin{vmatrix} 1 & 3 \\ 2 & k \end{vmatrix} = 1(k) - 3(2) = k - 6 = 0 \implies \mathbf{k} = 6$$

5. Data Science Application: Feature Multicollinearity Since $1 \text{ sq ft} = 0.092903 \text{ sq m}$, $x_2 = 0.092903x_1$. The linear combination $0.092903 \cdot x_1 - 1 \cdot x_2 + 0 \cdot x_3 = 0$ holds with non-zero coefficients. Thus the feature set $\{x_1, x_2, x_3\}$ is **linearly dependent**. In linear regression, this exact linear dependence leads to a singular matrix $X^T X$ ($\det(X^T X) = 0$), making $(X^T X)^{-1}$ undefined. One of the redundant size features must be dropped.

Week 4: Basis, Dimension & Matrix Rank

Detailed Solutions

1. Basis Verification & Coordinate Representation Form matrix $M = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. $\det(M) = 1(-1) - 1(1) = -2 \neq 0$. Since $\det(M) \neq 0$, $\mathcal{B}$ consists of 2 linearly independent vectors in $\mathbb{R}^2$. Therefore, $\mathcal{B}$ is a **valid basis for $\mathbb{R}^2$** .

To find coordinates c_1, c_2 for $\mathbf{v} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$:

$$\begin{cases} c_1 + c_2 = 5 \\ c_1 - c_2 = 1 \end{cases} \implies 2c_1 = 6 \implies \mathbf{c_1 = 3, c_2 = 2}$$

Vector $\mathbf{v} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

2. Finding a Basis for Subspace W From $x + 2y - z = 0 \implies z = x + 2y$. Any vector in W can be written as:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ x + 2y \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

Basis for W : $\mathcal{B}_W = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right\}$. Since there are 2 basis vectors, $\dim(\mathbf{W}) = 2$.

3. Matrix Rank Calculation

$$\begin{bmatrix} 1 & 2 & 1 & 4 \\ 2 & 4 & 3 & 11 \\ 3 & 6 & 4 & 15 \end{bmatrix} \xrightarrow[\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1}]{\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1}} \begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 1 & 3 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The REF has 2 non-zero rows (pivots in column 1 and column 3). Therefore, $\text{rank}(\mathbf{A}) = 2$.

4. Full Rank vs Rank Deficient

- **Maximum Possible Rank:** $\min(100, 5) = 5$.
- **Interpretation if $\text{rank}(X) = 3$:** The dataset is **rank deficient**. Out of 5 feature columns, only 3 are linearly independent; 2 features are redundant linear combinations of the others.

5. Data Science Application: Collinear Feature Removal Form matrix $F = [\mathbf{f}_1 \ \mathbf{f}_2 \ \mathbf{f}_3] = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 3 & 6 & 5 \end{bmatrix}$.

Row operations:

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 3 & 6 & 5 \end{bmatrix} \xrightarrow[\substack{R_3 \rightarrow R_3 - 3R_1}]{R_2 \rightarrow R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 5 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - 5R_2} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

- **Matrix Rank:** $\text{rank}(\mathbf{F}) = 2$.
- **Redundant Feature:** $\mathbf{f}_2 = 2\mathbf{f}_1$ is completely redundant.
- **Minimal Basis:** $\{\mathbf{f}_1, \mathbf{f}_3\} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 5 \end{bmatrix} \right\}$.

Week 5: Null Space, Nullity & Linear Mappings

Detailed Solutions

1. Null Space Calculation Solve $A\mathbf{x} = \mathbf{0}$ for $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & 5 \end{bmatrix}$:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 2 & 4 & 5 & 0 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 0 & 7 & 0 \end{array} \right]$$

- Row 2: $7x_3 = 0 \implies \mathbf{x}_3 = \mathbf{0}$.
- Row 1: $x_1 + 2x_2 - 0 = 0 \implies x_1 = -2x_2$.

Here $x_2 = t$ is a free variable.

$$\mathbf{x} = \begin{bmatrix} -2t \\ t \\ 0 \end{bmatrix} = t \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

- **Basis for Null(A):** $\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \right\}$.
- **Nullity:** $\text{nullity}(A) = \dim(\text{Null}(A)) = 1$.

2. Testing Linearity of Mappings

- **(a)** $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x - y \\ x + 3y \end{bmatrix}$: Notice that $T(\mathbf{x}) = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$. Since T is induced by matrix multiplication, it is a **valid Linear Transformation**.
- **(b)** $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x^2 \\ y \end{bmatrix}$: Test scalar multiplication for $c = 2$, $\mathbf{v} = [1, 1]^T$:

$$T(2\mathbf{v}) = T\left(\begin{bmatrix} 2 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 2^2 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$2T(\mathbf{v}) = 2T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

Since $\begin{bmatrix} 4 \\ 2 \end{bmatrix} \neq \begin{bmatrix} 2 \\ 2 \end{bmatrix}$, T fails scalar homogeneity and is **NOT a Linear Transformation**.

3. Matrix-Induced Transformation

$$T\left(\begin{bmatrix} 3 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ -2 & 3 \\ 5 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 1(3) + 0(-1) \\ -2(3) + 3(-1) \\ 5(3) + 4(-1) \end{bmatrix} = \begin{bmatrix} 3 \\ -9 \\ 11 \end{bmatrix}$$

4. Homogenous System Solutions For matrix $A \in \mathbb{R}^{3 \times 5}$:

- Domain dimension $n = 5$ (number of columns).
- By the Rank-Nullity Theorem: $\text{nullity}(A) = n - \text{rank}(A) = 5 - 3 = 2$.
- Since $\text{nullity}(A) = 2 > 0$, there exist **infinitely many non-trivial solutions** to $Ax = 0$.

5. Data Science Application: Linear Compression Null Space Matrix $W \in \mathbb{R}^{10 \times 50}$ maps inputs $\mathbf{x} \in \mathbb{R}^{50}$ ($n = 50$) to outputs $\mathbf{y} \in \mathbb{R}^{10}$. The maximum possible rank of W is $\min(10, 50) = 10$.

$$\text{nullity}(W) = n - \text{rank}(W) = 50 - \text{rank}(W) \geq 50 - 10 = 40$$

At least **40** dimensions of input variation are mapped to zero in the null space, representing the minimum lost information during linear compression.

Week 6: Matrix Representation, Kernel, Image & Rank-Nullity

Detailed Solutions

1. Matrix Representation Construction

$$T(\mathbf{e}_1) = T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 3(1) + 0 \\ 2(1) - 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$T(\mathbf{e}_2) = T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 + 1 \\ 0 - 4(1) \end{bmatrix} = \begin{bmatrix} 1 \\ -4 \end{bmatrix}$$

$$A = [T]_{\mathcal{E}} = \begin{bmatrix} 3 & 1 \\ 2 & -4 \end{bmatrix}$$

2. Non-Standard Basis Matrix Representation Basis $\mathcal{B} = \left\{ \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$.

- Apply T to $\mathbf{v}_1$: $T(\mathbf{v}_1) = \begin{bmatrix} 1 + 1 \\ 2(1) - 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

Express in basis $\mathcal{B}$: $c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \implies c_1 = 1, c_2 = 1 \implies [T(\mathbf{v}_1)]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

- Apply T to $\mathbf{v}_2$: $T(\mathbf{v}_2) = \begin{bmatrix} 1 + 0 \\ 2(1) - 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Express in basis $\mathcal{B}$: $c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \implies c_1 = 2, c_2 = -1 \implies [T(\mathbf{v}_2)]_{\mathcal{B}} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$. Matrix

Representation: $[T]_{\mathcal{B}} = \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}$.

3. Bases for Kernel and Image Row reduce $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow{\text{REF}} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$.

- **Pivots** are in Column 1 and Column 2.
- **(a) Basis for $\text{im}(A)$:** Take original columns 1 and 2 of A :

$$\mathcal{B}_{\text{im}} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix} \right\}, \quad \text{rank}(A) = 2$$

- **(b) Basis for $\ker(A)$:** Solve REF $A\mathbf{x} = \mathbf{0}$:
- Row 2: $-x_2 - 2x_3 = 0 \implies x_2 = -2x_3$.
- Row 1: $x_1 + 2(-2x_3) + 3x_3 = 0 \implies x_1 = x_3$.

Set $x_3 = t$: $\mathbf{x} = t \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$.

$$\mathcal{B}_{\ker} = \left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \right\}, \quad \text{nullity}(A) = 1$$

4. Rank-Nullity Verification Domain $V = \mathbb{R}^3$, so $\dim(V) = 3$.

$$\text{nullity}(A) + \text{rank}(A) = 1 + 2 = \mathbf{3} = \dim(\mathbb{R}^3) \quad \checkmark$$

5. Data Science Application: Linear Autoencoder Information Loss Given $W \in \mathbb{R}^{20 \times 100}$ and $\text{rank}(W) = 18$:

- **(a) Code Space Dimension:** $\dim(\text{im}(W)) = \text{rank}(W) = \mathbf{18}$.
- **(b) Lost Feature Dimensions:** By Rank-Nullity Theorem:

$$\dim(\ker(W)) = 100 - \text{rank}(W) = 100 - 18 = \mathbf{82} \text{ lost feature dimensions}$$

Week 7: Matrix Similarity, Affine Spaces, Norms & Inner Products

Detailed Solutions

1. Matrix Similarity Check

- **(a) Determinants:** $\det(A) = 1(3) - 0 = \mathbf{3}$, $\det(B) = 3(1) - 0 = \mathbf{3}$.
- **(b) Traces:** $\text{tr}(A) = 1 + 3 = \mathbf{4}$, $\text{tr}(B) = 3 + 1 = \mathbf{4}$.
- **(c) Similarity:** The eigenvalues of upper triangular A are $\lambda_1 = 1, \lambda_2 = 3$. The eigenvalues of lower triangular B are $\lambda_1 = 3, \lambda_2 = 1$. Since both have distinct eigenvalues $\{1, 3\}$, both matrices are diagonalizable to $D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$. Therefore, **A and B are SIMILAR** ($A \sim B$).

2. Weighted Inner Product

- **(a) Inner Product:**

$$\left\langle \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \end{bmatrix} \right\rangle_Q = 2(1)(3) + 5(2)(-1) = 6 - 10 = \mathbf{-4}$$

- **(b) Induced Norm:**

$$\left\| \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\|_Q = \sqrt{2(1)^2 + 5(2)^2} = \sqrt{2 + 20} = \sqrt{\mathbf{22}}$$

3. Cosine Similarity

$$\mathbf{d}_1 \cdot \mathbf{d}_2 = (1)(0) + (2)(1) + (0)(1) + (2)(1) = 0 + 2 + 0 + 2 = 4$$

$$\|\mathbf{d}_1\| = \sqrt{1^2 + 2^2 + 0^2 + 2^2} = \sqrt{9} = 3$$

$$\|\mathbf{d}_2\| = \sqrt{0^2 + 1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$\cos(\theta) = \frac{4}{3\sqrt{3}} \approx \mathbf{0.7698}$$

4. Cauchy-Schwarz Verification

$$\mathbf{u} \cdot \mathbf{v} = 3(1) + 4(2) = 11 \implies |\mathbf{u} \cdot \mathbf{v}| = 11$$

$$\|\mathbf{u}\| \|\mathbf{v}\| = \left(\sqrt{3^2 + 4^2} \right) \left(\sqrt{1^2 + 2^2} \right) = 5\sqrt{5} \approx \mathbf{11.1803}$$

Since $11 \leq 11.1803$, the **Cauchy-Schwarz inequality** $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$ holds true.

5. Data Science Application: L_1 vs L_2 Norm Regularization

- (a) L_1 Norm (Lasso):

$$\|\mathbf{w}\|_1 = |3| + |-4| + |0| + |12| = 3 + 4 + 0 + 12 = \mathbf{19}$$

- (b) L_2 Norm (Ridge):

$$\|\mathbf{w}\|_2 = \sqrt{3^2 + (-4)^2 + 0^2 + 12^2} = \sqrt{9 + 16 + 0 + 144} = \sqrt{169} = \mathbf{13}$$

Week 8: Orthogonality, Gram-Schmidt Process & Projections**Detailed Solutions****1. Projection onto a Vector**

$$\text{proj}_{\mathbf{u}}(\mathbf{v}) = \frac{\mathbf{v} \cdot \mathbf{u}}{\|\mathbf{u}\|^2} \mathbf{u} = \frac{4(1) + 3(2)}{1^2 + 2^2} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \frac{10}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

2. Gram-Schmidt Process (2×2)

$$1. \text{ Step 1: } \mathbf{u}_1 = \mathbf{v}_1 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}. \|\mathbf{u}_1\| = \sqrt{3^2 + 4^2} = 5.$$

$$\mathbf{q}_1 = \frac{\mathbf{u}_1}{5} = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$$

$$2. \text{ Step 2: } \mathbf{u}_2 = \mathbf{v}_2 - (\mathbf{v}_2 \cdot \mathbf{q}_1) \mathbf{q}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \left(1 \cdot \frac{3}{5} + 0\right) \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix} = \begin{bmatrix} 1 - 9/25 \\ -12/25 \end{bmatrix} = \begin{bmatrix} 16/25 \\ -12/25 \end{bmatrix}. \|\mathbf{u}_2\| = \sqrt{(16/25)^2 + (-12/25)^2} = \frac{20}{25} = \frac{4}{5}.$$

$$\mathbf{q}_2 = \frac{\mathbf{u}_2}{4/5} = \begin{bmatrix} 4/5 \\ -3/5 \end{bmatrix}$$

Orthonormal Basis: $\left\{ \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}, \begin{bmatrix} 4/5 \\ -3/5 \end{bmatrix} \right\}.$

3. Gram-Schmidt Process (3×3)

$$\bullet \mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \|\mathbf{u}_1\| = \sqrt{2} \Rightarrow \mathbf{q}_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}.$$

$$\bullet \mathbf{u}_2 = \mathbf{v}_2 - (\mathbf{v}_2 \cdot \mathbf{q}_1)\mathbf{q}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} = \begin{bmatrix} 1/2 \\ -1/2 \\ 1 \end{bmatrix}.$$

$$\|\mathbf{u}_2\| = \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{3}{2}} \Rightarrow \mathbf{q}_2 = \begin{bmatrix} 1/\sqrt{6} \\ -1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}.$$

$$\bullet \text{Repeating for } \mathbf{u}_3 \Rightarrow \mathbf{q}_3 = \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ -1/\sqrt{3} \end{bmatrix}.$$

4. Orthogonal Matrix Properties

- (a) $Q^T Q$ Computation:

$$Q^T Q = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \cos^2 \theta + \sin^2 \theta & 0 \\ 0 & \sin^2 \theta + \cos^2 \theta \end{bmatrix} = \mathbf{I}$$

- (b) Norm Preservation:

$$\|Q\mathbf{x}\|^2 = (Q\mathbf{x})^T(Q\mathbf{x}) = \mathbf{x}^T(Q^T Q)\mathbf{x} = \mathbf{x}^T \mathbf{I} \mathbf{x} = \mathbf{x}^T \mathbf{x} = \|\mathbf{x}\|^2 \Rightarrow \|Q\mathbf{x}\| = \|\mathbf{x}\|$$

5. Data Science Application: Projection Matrix Properties Given $P = X(X^T X)^{-1} X^T$:

- Symmetry ($P^T = P$):

$$P^T = (X(X^T X)^{-1} X^T)^T = (X^T)^T ((X^T X)^{-1})^T X^T = X(X^T X)^{-1} X^T = P$$

- Idempotence ($P^2 = P$):

$$P^2 = PP = X(X^T X)^{-1} \underbrace{(X^T X)(X^T X)^{-1}}_I X^T = X(X^T X)^{-1} X^T = P$$

Week 9: Multivariable Functions, Partial Derivatives & Gradient

Detailed Solutions

1. First-Order Partial Derivatives For $f(x, y) = x^2 y^3 + e^{2x} \sin(y)$:

$$\frac{\partial f}{\partial x} = 2xy^3 + 2e^{2x} \sin(y)$$

$$\frac{\partial f}{\partial y} = 3x^2 y^2 + e^{2x} \cos(y)$$

2. Gradient Vector Evaluation For $f(x, y) = 4x^2 y - 3y^2 + 5x$:

$$\frac{\partial f}{\partial x} = 8xy + 5 \Rightarrow f_x(1, 2) = 8(1)(2) + 5 = 16 + 5 = 21$$

$$\frac{\partial f}{\partial y} = 4x^2 - 6y \Rightarrow f_y(1, 2) = 4(1)^2 - 6(2) = 4 - 12 = -8$$

$$\nabla f(1, 2) = \begin{bmatrix} 21 \\ -8 \end{bmatrix}$$

3. Directional Derivative 1. Gradient $\nabla f(x, y) = \begin{bmatrix} 2x + 3y \\ 3x + 2y \end{bmatrix}$. At $(2, 1)$: $\nabla f(2, 1) = \begin{bmatrix} 2(2) + 3(1) \\ 3(2) + 2(1) \end{bmatrix} = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$. 2. Normalize direction vector $\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \implies \mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$. 3. Directional derivative:

$$D_{\mathbf{u}}f(2, 1) = \nabla f(2, 1) \cdot \mathbf{u} = \begin{bmatrix} 7 \\ 8 \end{bmatrix} \cdot \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix} = \frac{21 + 32}{5} = \frac{53}{5} = 10.6$$

4. Steepest Ascent Direction For $f(x, y) = x^3y - 2xy^2$:

- $f_x = 3x^2y - 2y^2 \implies f_x(1, -1) = 3(1)(-1) - 2(1) = -5$.
- $f_y = x^3 - 4xy \implies f_y(1, -1) = 1 - 4(1)(-1) = 1 + 4 = 5$.

Gradient: $\nabla f(1, -1) = \begin{bmatrix} -5 \\ 5 \end{bmatrix}$.

- **Unit Direction of Steepest Ascent:** $\mathbf{u} = \frac{\nabla f}{\|\nabla f\|} = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$.
- **Maximum Rate of Increase:** $\|\nabla f(1, -1)\| = \sqrt{(-5)^2 + 5^2} = \sqrt{50} = 5\sqrt{2} \approx 7.071$.

5. Data Science Application: MSE Loss Gradient For $L(m, b) = (mx_i + b - y_i)^2$: Using the chain rule:

$$\frac{\partial L}{\partial m} = 2(mx_i + b - y_i) \cdot x_i = 2x_i(mx_i + b - y_i)$$

$$\frac{\partial L}{\partial b} = 2(mx_i + b - y_i) \cdot 1 = 2(mx_i + b - y_i)$$

$$\nabla L(m, b) = \begin{bmatrix} 2x_i(mx_i + b - y_i) \\ 2(mx_i + b - y_i) \end{bmatrix}$$

Week 10: Tangent Planes, Steepest Descent & Critical Points

Detailed Solutions

1. Tangent Plane Equation For $z = x^2 + 3y^2 - 4x$ at $(x_0, y_0) = (2, 1)$:

- $z_0 = f(2, 1) = 2^2 + 3(1)^2 - 4(2) = 4 + 3 - 8 = -1$.
- $f_x = 2x - 4 \implies f_x(2, 1) = 2(2) - 4 = 0$.
- $f_y = 6y \implies f_y(2, 1) = 6(1) = 6$.

Tangent Plane:

$$z - (-1) = 0(x - 2) + 6(y - 1) \implies z + 1 = 6y - 6 \implies \mathbf{z = 6y - 7}$$

2. Gradient Descent Iteration Step

- **(a) Gradient:** $\nabla L(w_1, w_2) = \begin{bmatrix} 2w_1 \\ 8w_2 \end{bmatrix}$.

- **(b) Update Step:** At $\mathbf{w}^{(0)} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$: $\nabla L(3, 2) = \begin{bmatrix} 6 \\ 16 \end{bmatrix}$.

$$\mathbf{w}^{(1)} = \mathbf{w}^{(0)} - \alpha \nabla L = \begin{bmatrix} 3 \\ 2 \end{bmatrix} - 0.1 \begin{bmatrix} 6 \\ 16 \end{bmatrix} = \begin{bmatrix} 3 - 0.6 \\ 2 - 1.6 \end{bmatrix} = \begin{bmatrix} 2.4 \\ 0.4 \end{bmatrix}$$

3. Finding Critical Points (2×2) Set first partial derivatives of $f(x, y) = x^3 + y^3 - 3xy$ to zero:

1. $f_x = 3x^2 - 3y = 0 \implies y = x^2$.
2. $f_y = 3y^2 - 3x = 0 \implies x = y^2$.

Substitute $y = x^2$ into $x = y^2$:

$$x = (x^2)^2 = x^4 \implies x^4 - x = 0 \implies x(x^3 - 1) = 0$$

- If $x = 0 \implies y = 0^2 = 0 \implies$ Critical point $(0, 0)$.
- If $x = 1 \implies y = 1^2 = 1 \implies$ Critical point $(1, 1)$.

4. Linear Approximation For $f(x, y) = \sqrt{x^2 + y^2}$ at $(3, 4)$: $f(3, 4) = 5$.

- $f_x = \frac{x}{\sqrt{x^2 + y^2}} \implies f_x(3, 4) = \frac{3}{5} = 0.6$.
- $f_y = \frac{y}{\sqrt{x^2 + y^2}} \implies f_y(3, 4) = \frac{4}{5} = 0.8$.

Linear approximation function: $L(x, y) = 5 + 0.6(x - 3) + 0.8(y - 4)$. For $(3.06, 3.92)$:

$$L(3.06, 3.92) = 5 + 0.6(0.06) + 0.8(-0.08) = 5 + 0.036 - 0.064 = \mathbf{4.972}$$

5. Data Science Application: Least Squares Gradient Descent Update For $L(w) = (2w - 6)^2$:

$$\frac{dL}{dw} = 2(2w - 6) \cdot 2 = 8w - 24$$

- **Step 1:** At $w^{(0)} = 0$, $\frac{dL}{dw} = -24 \implies w^{(1)} = 0 - 0.05(-24) = \mathbf{1.2}$.
- **Step 2:** At $w^{(1)} = 1.2$, $\frac{dL}{dw} = 8(1.2) - 24 = -14.4 \implies w^{(2)} = 1.2 - 0.05(-14.4) = 1.2 + 0.72 = \mathbf{1.92}$.

The weight sequence $0 \rightarrow 1.2 \rightarrow 1.92$ rapidly converges toward the optimal minimum $w^* = 3$.

Week 11: Hessian Matrix, Multivariable Extrema & Differentiability

Detailed Solutions

1. Hessian Matrix Computation For $f(x, y) = x^3 - 3xy^2 + y^4$:

- First partials: $f_x = 3x^2 - 3y^2$, $f_y = -6xy + 4y^3$.
- Second partials: $f_{xx} = 6x$, $f_{xy} = -6y$, $f_{yy} = -6x + 12y^2$.

$$H(f)(x, y) = \begin{bmatrix} 6x & -6y \\ -6y & -6x + 12y^2 \end{bmatrix}$$

At point $(1, 2)$:

$$H(f)(1, 2) = \begin{bmatrix} 6(1) & -6(2) \\ -6(2) & -6(1) + 12(2^2) \end{bmatrix} = \begin{bmatrix} 6 & -12 \\ -12 & 42 \end{bmatrix}$$

2. Classifying 2D Critical Points For $f(x, y) = x^3 + y^3 - 3xy$, second partials are $f_{xx} = 6x$, $f_{xy} = -3$, $f_{yy} = 6y$. Hessian Discriminant: $D(x, y) = f_{xx}f_{yy} - (f_{xy})^2 = (6x)(6y) - (-3)^2 = 36xy - 9$.

- **At (0, 0):** $D(0, 0) = 36(0) - 9 = -9 < 0$. Therefore, (0, 0) is a **Saddle Point**.
- **At (1, 1):** $D(1, 1) = 36(1)(1) - 9 = 27 > 0$, and $f_{xx}(1, 1) = 6 > 0$. Therefore, (1, 1) is a **Local Minimum**.

3. Saddle Point Identification For $f(x, y) = x^2 - y^2$ at critical point (0, 0):

$$H(f)(0, 0) = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$$

Determinant $D = 2(-2) - 0 = -4 < 0$. Since $D < 0$, the Hessian is indefinite with eigenvalues $\lambda_1 = 2 > 0$ and $\lambda_2 = -2 < 0$. Thus (0, 0) is a **Saddle Point**.

4. Classifying 3D Critical Points For $f(x, y, z) = x^2 + 2y^2 + 3z^2 - 4x + 8y - 12z$:

- First partials: $f_x = 2x - 4 = 0 \implies x = 2$, $f_y = 4y + 8 = 0 \implies y = -2$, $f_z = 6z - 12 = 0 \implies z = 2$.

Critical point: **(2, -2, 2)**.

- Hessian matrix:

$$H(f)(2, -2, 2) = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

All diagonal entries (eigenvalues) are strictly positive (2, 4, 6 > 0), meaning $H(f)$ is positive-definite. Therefore, (2, -2, 2) is a **Local Minimum**.

5. Data Science Application: Convexity of Mean Squared Error For $L(w) = \frac{1}{N} \sum_{i=1}^N (wx_i - y_i)^2$:

- First derivative: $\frac{dL}{dw} = \frac{2}{N} \sum_{i=1}^N x_i (wx_i - y_i)$.
- Second derivative: $\frac{d^2L}{dw^2} = \frac{2}{N} \sum_{i=1}^N x_i^2$.

Since $x_i^2 \geq 0$ for all real sample features x_i , the second derivative $\frac{d^2L}{dw^2} \geq 0$ for all w . Because the second derivative is non-negative everywhere, $L(w)$ is **strictly convex**, ensuring that any local minimum found by Gradient Descent is the **unique global minimum**.

Week 12: Eigenvalues, Eigenvectors, Diagonalization & SVD

Detailed Solutions

1. Eigenvalues & Eigenvectors (2×2)

$$\det(A - \lambda I) = \begin{vmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{vmatrix} = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2) = 0$$

- Eigenvalues: $\lambda_1 = 5$, $\lambda_2 = 2$.

- **Eigenvector for $\lambda_1 = 5$:** $(A - 5I)\mathbf{v}_1 = \mathbf{0} \implies \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{0} \implies x = y \implies \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

- **Eigenvector for $\lambda_2 = 2$:** $(A - 2I)\mathbf{v}_2 = \mathbf{0} \implies \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{0} \implies 2x + y = 0 \implies y = -2x \implies \mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$.

2. Matrix Diagonalization & Matrix Powers

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}, \quad P^{-1} = \frac{1}{-3} \begin{bmatrix} -2 & -1 \\ -1 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$$

To compute $A^4 = PD^4P^{-1}$:

$$D^4 = \begin{bmatrix} 5^4 & 0 \\ 0 & 2^4 \end{bmatrix} = \begin{bmatrix} 625 & 0 \\ 0 & 16 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 625 & 0 \\ 0 & 16 \end{bmatrix} \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 625 & 16 \\ 625 & -32 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$$

$$A^4 = \frac{1}{3} \begin{bmatrix} 1266 & 609 \\ 1218 & 657 \end{bmatrix} = \begin{bmatrix} 422 & 203 \\ 406 & 219 \end{bmatrix}$$

3. Symmetric Matrix Spectral Decomposition For $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$: $\det(A - \lambda I) = (3 - \lambda)^2 - 1 = 0 \implies (3 - \lambda) = \pm 1 \implies \lambda_1 = 4, \lambda_2 = 2$.

- For $\lambda_1 = 4$: Normalized unit eigenvector $\mathbf{q}_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$.
- For $\lambda_2 = 2$: Normalized unit eigenvector $\mathbf{q}_2 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$.

Orthogonal Matrix $Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ satisfies $Q^T Q = I$ and $A = Q D Q^T$.

4. Singular Values Calculation For $A = \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}$:

$$A^T A = \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} 9 & 0 \\ 0 & 4 \end{bmatrix}$$

Eigenvalues of $A^T A$ are $\lambda_1 = 9, \lambda_2 = 4$. Singular Values: $\sigma_1 = \sqrt{9} = 3, \sigma_2 = \sqrt{4} = 2$.

5. Comprehensive Course Review: PCA via SVD

- **(a) Why Principal Components are Eigenvectors of C :** The covariance matrix $C = \frac{1}{N} X^T X$ measures feature variances and co-variances. The eigenvectors of C define orthogonal axes along which feature variance is maximized, decorrelating features.
- **(b) Percentage of Variance Explained:**

$$\text{Total Variance} = \lambda_1 + \lambda_2 + \lambda_3 = 80 + 15 + 5 = 100$$

$$\text{Explained Variance by PC1 \& PC2} = \frac{\lambda_1 + \lambda_2}{\text{Total}} = \frac{80 + 15}{100} = \frac{95}{100} = 95\%$$